# APEX INTEGRATION ENGINEERING SQUAD

Enterprise Levelpath, ERP (Oracle/SAP) & CLM Integration Specialists

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Live Portfolio: viranga-lakshan.github.io/portfolio/

TARGET CLIENT: WONDER SERVICES | Role: Levelpath Integration Engineering Team

## Technical Integration Engineering Proposal: Levelpath, ERP & CLM Rollout

A dedicated, multi-disciplinary engineering squad providing end-to-end integration architecture, API development, data transformation, and agile project governance for Wonder Services' enterprise Levelpath implementation. Dedicated full-time capacity across the 12-month engagement with zero single-point-of-failure.

| 40+ Hrs/Wk | 3 Engineers | REST & OAuth 2.0 | S2P & P2P |
|---|---|---|---|
| FULL-TIME DEDICATED CAPACITY | SQUAD COVERAGE & REDUNDANCY | API SECURITY & MIDDLEWARE | PROCUREMENT & ERP DOMAIN |

### The Integration Squad Advantage for Wonder Services

Enterprise Levelpath rollouts require a rare combination of capabilities: deep hands-on REST API / OAuth 2.0 development, low-level ERP data schema mapping (Oracle Fusion / SAP), and rigorous stakeholder/workflow coordination. While Wonder Services budgeted for an individual integration engineer, **we present a specialized 3-person Integration Squad working under a unified contractor engagement**.

By engaging our team, Wonder Services gains a **Lead Architect** (Viranga Lakshan), a **Senior Backend & Oracle SQL Integration Engineer** (Janith Nadiranga), and a **Procurement Systems Project Manager / BA** (Lithumitha Ravikumar). You achieve complete delivery redundancy, accelerated integration turnaround, and professional agile documentation—at the cost of a single senior contractor.

### Alignment with Wonder Services Key Requirements

| WONDER SERVICES REQUIREMENT | SQUAD EXECUTION STRATEGY & ASSIGNED OWNERSHIP |
|---|---|
| **Levelpath / Procurement Workflows** | Configure Levelpath intake forms, conditional routing rules (spend threshold, cost center), approval chains, and supplier onboarding pipelines. *(Lead: Lithumitha & Viranga)* |
| **ERP & CLM Connectivity (Oracle/Ironclad)** | Build robust REST interfaces and OAuth 2.0 connectors linking Levelpath to Oracle Cloud ERP / SAP and Ironclad CLM for bi-directional contract and PO synchronization. *(Lead: Viranga & Janith)* |
| **Data Mapping, Transformation & SQL** | Manage complex JSON payload normalization, Oracle SQL schemas, ETL scripts, and 3-way matching validation rules across purchase requisitions and invoices. *(Lead: Janith)* |
| **Production Support, Logs & Go-Live** | Diagnose integration dropouts via ELK / CloudWatch logs, implement idempotent retries with dead-letter queues, and execute sprint releases with technical documentation. *(Lead: Squad)* |

## Meet the Integration Squad: Roles, Credentials & Responsibilities

### VIRANGA LAKSHAN
**Lead Integration Architect & ERP Engineer**

Senior Software Engineer with 5+ years of enterprise experience at Arimac Digital. Specializes in enterprise ERP integrations, REST API gateways, and OAuth 2.0 security.

- **ERP Integrations:** Architected SAP ERP integration gateway for DIMO Lanka vehicle logistics platform.
- **API & Security:** OAuth 2.0 / JWT, Keycloak, mTLS, rate limiting, and token refresh managers.
- **Event Streaming:** Kafka audit event pipelines capturing procurement state changes.
- **Tech Stack:** Java 21, Spring Boot, REST APIs, JSON, Docker, CI/CD, AWS S3.

### JANITH NADIRANGA
**Backend Integration & Oracle SQL Specialist**

Software Engineer at DMS Electronics with deep expertise in Java/Spring Boot, Oracle SQL, PostgreSQL, AP reconciliation, and high-volume data pipelines.

- **3-Way Matching Engine:** Built *InvoiceIQ* accounts payable reconciliation REST API with deterministic 3-way matching.
- **Procurement Data:** Engineered *Nexware* warehouse & PO system with Kafka domain events.
- **Databases:** Advanced Oracle SQL, PostgreSQL, schema migrations, and indexing.
- **Tech Stack:** Java, Spring Boot 3, Oracle SQL, Python/FastAPI, Docker, RabbitMQ.

### LITHUMITHA RAVIKUMAR
**Integration Project Manager & S2P Systems Analyst**

Associate Project Manager at Arimac Digital. First Class Honours in Business Information Systems. Specializes in agile delivery and client stakeholder management.

- **Procurement Workflows:** Gathers S2P/P2P business rules and writes user stories & acceptance criteria.
- **Stakeholder Liaison:** Bridges business procurement users, legal teams, and technical engineers.
- **Release Management:** Manages sprint planning, JIRA backlogs, UAT tracking, and go-live signoffs.
- **Tooling:** JIRA, Confluence, MS Project, Agile Scrum/Waterfall, SRS authoring.

## Squad Technical Capabilities & Architecture Coverage Matrix

| INTEGRATION DOMAIN | SQUAD TOOLS & TECHNOLOGIES | WONDER SERVICES PROJECT IMPACT |
|---|---|---|
| **API Protocols & Security** | RESTful APIs, JSON, XML, OAuth 2.0 (Client Credentials, JWT), SSO, Webhooks, mTLS | Enforces bulletproof security between Levelpath, middleware, ERP, and Ironclad CLM. |
| **Enterprise ERP & Databases** | Oracle Cloud ERP / Fusion patterns, SAP ERP, Oracle SQL, PostgreSQL, MySQL | Direct experience querying and synchronizing vendor master, POs, and GL accounts. |
| **Backend & Middleware** | Java 21, Spring Boot 3, Python (FastAPI), Kafka, RabbitMQ, Docker, Linux | High-throughput data transformation pipelines, event streaming, and custom middleware handlers. |
| **S2P / P2P Domain Logic** | Intake forms, Dynamic Routing, Requisitions, 3-Way Matching, Supplier Onboarding | Zero ramp-up time on procurement business logic; immediate grasp of client workflows. |
| **Agile & Release Delivery** | JIRA, Confluence, Git, GitHub Actions, Jenkins, Postman, OpenAPI/Swagger | Predictable sprint velocity, daily status reporting, comprehensive API documentation. |

## Procure-to-Pay (P2P) & Source-to-Pay (S2P) Process Mastery

Our squad operates with native fluency across the entire procurement lifecycle: (1) **Intake & Requisition:** Smart routing in Levelpath based on category codes, spend ceilings, and risk factors; (2) **Supplier Onboarding:** W-9 compliance, banking verification, and ERP vendor master deduplication; (3) **Contracting (CLM):** Synchronization of contract status, signed PDFs, and spend commitments with Ironclad; (4) **PO Creation:** Automated ERP purchase order generation; (5) **Invoice & 3-Way Matching:** Validation of PO lines, receipts, and invoice tolerances to eliminate payment leakage.

## Proven Enterprise Projects & Integration Case Studies

### DIMO Enterprise Logistics & ERP Integration Platform (SAP ERP Gateway & Microservices)

Lead: Viranga Lakshan · Arimac Digital

Architected an enterprise dispatch, logistics, and multi-tier approval platform for premier conglomerate DIMO Lanka, connecting directly with SAP ERP and managing complex organizational approval chains.

- ✓ Engineered REST API gateways communicating with **SAP ERP**, enforcing OAuth 2.0 / JWT authentication and audit-compliant transactional logging.
- ✓ Implemented **Kafka event streaming** for real-time state transitions across requisition, approval, and dispatch lifecycles.
- ✓ Secured enterprise document upload pipelines to AWS S3 with automated ClamAV antivirus scanning and Azure AD identity federation.

SAP ERP Integration | Java 21 / Spring Boot | OAuth 2.0 / JWT | REST APIs & JSON | Spring Kafka | Docker CI/CD

### InvoiceIQ — Accounts Payable Reconciliation & 3-Way Matching Engine

Lead: Janith Nadiranga · Portfolio

Architected a deterministic accounts payable reconciliation REST API platform designed to automate invoice verification, three-way matching against purchase orders, and exception management.

- ✓ Implemented deterministic **3-way matching algorithms** with high-precision decimal calculations to detect price variances, quantity discrepancies, and duplicate vendor billings.
- ✓ Engineered immutable human-in-the-loop approval workflows with SHA-256 idempotency controls and append-only audit logging.
- ✓ Built high-performance REST APIs using Python/FastAPI and integrated external document parsing pipelines.

3-Way Matching | Accounts Payable (AP) | REST APIs | Oracle SQL / PostgreSQL | Idempotency Controls

### Nexware — Warehouse, Purchase Order & Supplier Management Platform

Lead: Janith Nadiranga · Portfolio

Full-stack modular enterprise application managing end-to-end procurement, purchase orders, inventory reservation, and supplier lifecycle workflows across multi-tier distribution networks.

- ✓ Developed modular REST APIs with JWT authentication, role-based access control, and Kafka event handling across inventory, PO, and supplier domains.
- ✓ Engineered complex PostgreSQL / Oracle SQL database schemas supporting high-volume transactional throughput and audit traceability.

Java 21 / Spring Boot 3 | Purchase Orders (PO) | Supplier Lifecycle | Kafka Events | Docker CI/CD

### Enterprise Analytics, SaaS & Client Dashboard Deliveries

Lead: Lithumitha Ravikumar · Arimac Digital

Managed end-to-end delivery of enterprise dashboard and SaaS products for international clients, driving requirements elicitation, sprint execution, and release management across distributed engineering teams.

- ✓ Authored comprehensive Software Requirement Specifications (SRS), user stories, and acceptance criteria for enterprise analytics platforms.
- ✓ Coordinated sprint ceremonies in JIRA, managed dependencies and risks, and ensured seamless communication between client stakeholders and engineering.

Agile Scrum / JIRA | Requirements Elicitation | Stakeholder Management | Release Validation

## Technical Execution Blueprint: Levelpath ↔ Middleware ↔ ERP / CLM Pipeline

### 1. Intake Routing & Schema Mapping

Configure Levelpath's dynamic intake forms to normalize incoming requisitions, map category codes, and transform JSON payloads to ERP line-item specifications.

### 2. Secure OAuth 2.0 & Token Caching

Implement client credentials authentication with thread-safe token caching and proactive pre-expiration refresh to prevent 401 mid-flight drops.

### 3. Bi-Directional Master Data Synchronization

Keep ERP vendor masters, chart of accounts (COA), and cost centers synchronized with Levelpath; push approved purchase requisitions and contracts to ERP/Ironclad.

### 4. Resilient Retries & Idempotency

Enforce unique correlation IDs (`X-Correlation-ID`), exponential backoff for transient 5xx errors, and dead-letter queue alerting for payload schema drift.

## Production Troubleshooting Case Study: Token Race Conditions & Schema Drift

**Root Cause Analysis:** In an enterprise ERP dispatch pipeline, asynchronous worker threads shared an OAuth 2.0 Bearer token that expired mid-flight during high-volume spikes. Outgoing requests to the ERP endpoint failed with 401s before token refresh completed, triggering cascading retries that starved connection pools.
**Engineering Remedy:** Re-architected the token manager with a proactive 5-minute pre-expiration refresh buffer and distributed lock. Wrapped all outbound calls in an idempotent HTTP client with exponential backoff and correlation IDs. Result: 100% elimination of 401 dropouts, zero duplicate postings, and a 34% drop in API latency overhead.

### ⏱️ Capacity & Engagement Commitment

- **Dedicated Capacity:** 40+ Hours / Week dedicated across the squad.
- **Contract Duration:** Ready for full 12-month engagement.
- **Time Zone Overlap:** Daily overlap with US (EST/CST) and UK/European business hours.

### 🤝 Collaboration & Working Model

- **Immediate Kickoff:** Ready to commence immediately upon award.
- **Agile Process:** Daily Slack / Teams communication, sprint reviews, and JIRA backlog grooming.
- **Commercial Model:** Simple single-contract hourly engagement—you pay standard contractor rates for 3-person squad output.

**Ready to Deploy for Wonder Services' Levelpath Implementation**
Available for an introductory technical call with Wonder Services leadership to review integration architecture.

⚡ Full Squad Readiness (40+ Hrs/Wk)